

Migraine Trigger Ranking & Prediction

A Preventive Machine Learning Approach

mHealth Diary & Environmental Context Focus

Project Presentation

Digital health assistance using machine learning for advanced care.

Focus: 24–48 hour pre-attack forecasting and personalized trigger ranking

Executive Summary & Value Proposition

Clinical Need: Migraine causes substantial disability and affects quality of life and workplace productivity.

Technological Shift: Move from reactive post-attack care toward proactive 24–48 hour risk forecasting.

Key Innovation: Combine mHealth diary entries, premonitory symptoms, and environmental context such as weather and barometric pressure.

Personalization: Rank the factors most strongly associated with an individual's upcoming attack.

Expected Value: Support timely behavioral action and informed clinical decision-making using an everyday smartphone.

Problem Statement — Trigger Stacking & Causality

Trigger stacking: Small stressors such as sleep changes, stress, caffeine/alcohol shifts, and pressure changes may accumulate.

Delayed effects: Trigger exposure may occur 24–48 hours before severe headache onset, making causal recall difficult.

Premonitory confounding: Fatigue, yawning, cravings, or neck stiffness may be early warning signs rather than causes.

Data gap: Conventional diaries often emphasize attack history; forecasting requires structured pre-attack observations.

Research Goal: Learn temporal relationships between daily context and the probability of an upcoming migraine.

Sustainable Development Goal Alignment

Primary SDG: Goal 3 — Good Health & Well-Being.

Target 3.4: Contribute to reducing disability from non-communicable diseases and improving well-being.

Accessibility: Smartphone-first design avoids dependence on expensive wearable hardware.

Patient Empowerment: Converts subjective recall into transparent, data-supported insights.

Scalability: A software-based approach can potentially reach diverse and resource-constrained populations.

Dataset & Multi-Stream Feature Architecture

mHealth Diary: Sleep duration/quality, stress score, physical activity, caffeine/alcohol, and premonitory symptoms.

Historical Headache Data: Previous-day pain intensity, duration, and medication use.

Environmental Context: Temperature, barometric pressure, humidity, and Air Quality Index (AQI).

Temporal Features: Lagged observations such as $t-1$ and $t-2$ capture changes preceding an attack.

Feature Selection: Statistical testing and feature-selection methods can reduce redundancy and improve model stability.

Machine Learning Models & Algorithms

Random Forest: Handles mixed feature types and nonlinear interactions; provides feature-importance rankings.

Gradient Boosting / XGBoost: Sequentially reduces prediction error and performs strongly on structured data.

Artificial Neural Network: Learns complex nonlinear relationships across clinical and behavioral features.

Time-Series Modeling: Sliding 24–48 hour lag windows capture trajectories rather than isolated daily values.

Model Comparison: Evaluate several approaches and select the model that best balances predictive performance and interpretability.

Class Imbalance & Clinical Thresholding

Challenge: Migraine attack days may form a minority of diary records.

Class Weighting: Penalize missed attack days more strongly during training.

Threshold Calibration: Use a decision threshold chosen for clinically acceptable specificity.

False-Alarm Concern: Excessive alerts may encourage unnecessary medication use and reduce user trust.

Evaluation Metrics: ROC-AUC, F1-score, precision, recall, and Precision–Recall curves should be considered alongside accuracy.

System Architecture & Preventive Digital Health Solution



Low-burden UI: Short daily micro-surveys designed to reduce tracking fatigue.

Adaptive Personalization: Periodically update the model using new diary observations and feedback.

Explainable AI: SHAP-style explanations can show which recent factors contributed to elevated risk.

Output: A simple risk level plus ranked contributing factors, rather than an opaque prediction.

Clinical & Social Impact

Preemptive Intervention: A 24–48 hour warning may create time for hydration, rest, stress management, or clinician-approved early treatment.

Physician Decision Support: Summaries can highlight attack frequency, ranked factors, and lifestyle/environmental patterns.

Patient Engagement: Personalized explanations may improve understanding of individual patterns.

Cost Effectiveness: Smartphone-based delivery reduces hardware requirements.

Long-Term Goal: Support preventive migraine management while keeping the system practical and clinically responsible.

References

Das, S., & Hridy, R. I. (2025). Machine Learning-based Migraine Prediction: Analyzing Key Features and Cause-Effect Relationships. *International Journal of Computer Applications*, 187(11), 13–23.

Faisal, F., Poole, A. C., et al. (2026). Forecasting migraine with time-series machine learning from mobile health data. *The Journal of Headache and Pain*, 27(1), 91.

Mahajan, S., et al. (2024). AI-Powered Migraine Trigger Tracker. *IJRTI*, 9(6), 233–248.

Munson, S. A., Schroeder, J., et al. (2020). The Importance of Starting With Goals in N-of-1 Studies. *Frontiers in Digital Health*, 2, 3.

Reddy, A., & Reddy, A. (2025). Migraine triggers, phases, and classification using machine learning models. *Frontiers in Neurology*, 16, 1555215.

Stubberud, A., Ingvaldsen, S. H., et al. (2023). Forecasting migraine with machine learning based on mobile phone diary and wearable data. *Cephalalgia*, 43(5), 1–10.